

The Pattern Zoo

HackerRank C interview — recognition checklist. ~14 patterns cover the vast majority of screens. Recognition first; speed comes from use.

The 3-step reflex (under the clock):

1. **Read the bound on n** → that fixes the *allowed complexity* before you read the problem twice.
2. **Scan the cue column** → 2–3 patterns light up.
3. **The constraints kill the impossible ones** → usually one survives.

Pattern → Dead-giveaway cue

PATTERN	DEAD-GIVEAWAY CUE
1 Two-pointer	sorted (or sortable) array, “find a pair/triple”, converge from both ends
2 Sliding window	“longest/shortest contiguous subarray/substring with property”, no negatives
3 Prefix sum + hashmap	“contiguous sum = k”, especially with negatives (window is dead)
4 Hashmap / freq array	“count occurrences”, “seen before”, “distinct”, “anagram”
5 Sorting + sweep	“intervals/meetings overlap or merge”, “schedule”
6 Binary search (array)	sorted + “find/insert position of value”
7 Binary search on answer	“min/max value such that property holds”, “possible within budget K?” + monotonic
8 Heap (priority queue)	“kth largest/smallest”, “top k”, “running median”
9 Monotonic stack	“next greater/smaller element”, spans, histogram
10 Stack	“balanced”, “valid parentheses”, nesting, undo
11 BFS	“minimum steps to reach”, unweighted shortest path, levels
12 DFS / union-find	“connected components”, flood fill, “all paths”
13 Dynamic programming	“number of ways”, “min/max cost”, overlapping subproblems
14 Math / bit trick	huge n , divisibility/parity, “without extra space”

Constraint → complexity

Budget ~1e8 ops/sec. Work backwards from n .

N BOUND	ALLOWED TIME	LIKELY APPROACH
≤ 12	$O(n!)$ / $O(2^n \cdot n)$	brute force, backtracking
≤ 25	$O(2^n)$	subsets, bitmask DP, meet-in-middle
≤ 500	$O(n^3)$	Floyd-Warshall, interval DP
$\leq 5e3$	$O(n^2)$	2 nested loops, basic DP table
$\leq 1e6$	$O(n \log n)/O(n)$	sort, two-pointer, window, hashing, heap
$\leq 1e8$	$O(n)$	single pass, prefix sums, counting
$\leq 1e18$	$O(\log n)/O(1)$	binary search, math/closed form

When stuck — 3 questions

ASK	IF YES →
Can I sort it?	two-pointer, sweep, binary search
Can a hashmap make lookup $O(1)$?	freq count, “seen before”, prefix-sum
Can I keep a running quantity in one pass?	window, prefix sum, counting

C gotchas (the **X** on submit)

- `long long` for sums $> \sim 2e9$; `print %lld`.
- `mid = lo + (hi-lo)/2` — never `(lo+hi)/2`.
- Comparator `(a>b)-(a<b)`, not `a-b` (overflow).
- Off-by-one on 1-indexed input.
- Consume newline before `fgets` after `scanf("%d")`.